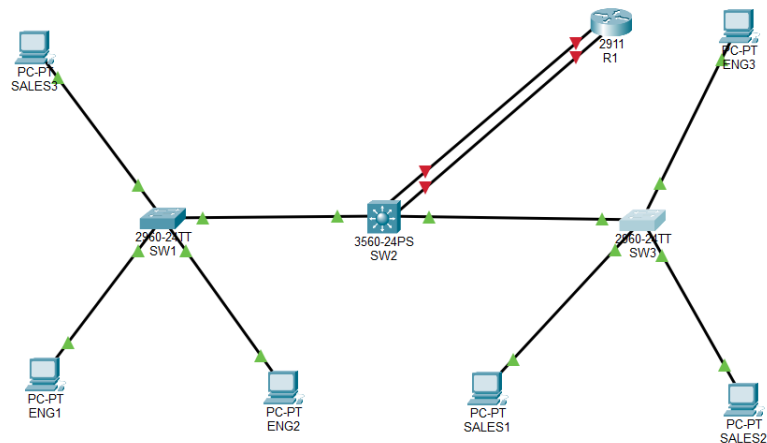


Week 9: VLAN and Inter-VLAN Routing Configuration (10 Marks)

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89)

```
Switch>ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vtp mode server
Device mode already VTP SERVER.
Switch(config)#vtp domain CAMPUS
Changing VTP domain name from NULL to CAMPUS
Switch(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to CAMPUS.

Switch(config)#vtp password cisco123
Setting device VLAN database password to cisco123
Switch(config)#vtp version 2
Switch(config)#
```

90)

```

Switch#enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name ENG
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name SALES
Switch(config-vlan)#exit
Switch(config)#do show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	ENG	active	
20	SALES	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```

Switch(config)#

```

91) SW1:

```

Switch>ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface g0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Switch(config-if)#switch port trunk allowed vlan 10,20
^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport trunk allowed vlan 10,20

```

SW2:

```

Switch#ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface g0/1
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk" mode.
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#exit
Switch(config)#interface g0/2
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk" mode.
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk" mode.
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#

```

SW3:

```
Switch#ena
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface g0/2
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#
```

92) SW1

```

Switch#show vlan brief
VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/2

1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active

Switch#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Gig0/1    10,20

Port      Vlans allowed and active in management domain
Gig0/1    none

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    none

Switch#show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          : CAMPUS
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 00D0.978A.4800
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00
Local updater ID is 0.0.0.0 (no valid interface found)

Feature VLAN :
-----
VTP Operating Mode       : Server
Maximum VLANs supported locally : 255
Number of existing VLANs : 5
Configuration Revision   : 0
MDS digest               : 0xEA 0x87 0x1F 0x3C 0x65 0x51 0x13 0x99
                           0xF1 0xE3 0x0D 0x72 0x54 0x44 0xA8 0x23

Switch#

```

SW2:

```
SW2
Physical Config CLI Attributes
IOS Command Line Interface
$SYS-5-CONFIG_1: Configured from console by console
Switch#show vlan brief
VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
10   ENG                    active
20   SALES                  active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    auto      n-802.1q       trunking    1
Gig0/2    auto      n-802.1q       trunking    1

Port      Vlans allowed on trunk
Gig0/1    10,20
Gig0/2    10,20

Port      Vlans allowed and active in management domain
Gig0/1    10,20
Gig0/2    10,20

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    10,20
Gig0/2    10,20

Switch#show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 2
VTP Domain Name          : CAMPUS
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 000A.F391.1000
Configuration last modified by 0.0.0.0 at 3-1-93 00:23:00
Local updater ID is 0.0.0.0 (no valid interface found)

Feature VLAN :
-----
VTP Operating Mode       : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs : 7
Configuration Revision   : 5
MDS digest               : 0x83 0x33 0xB3 0x7F 0x65 0xB7 0x32 0x76
                        : 0x11 0x48 0xE4 0x92 0xE4 0x6D 0xDB 0x3F
```

SW3:

```
Switch#show vlan brief
VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/2    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Gig0/2    10,20

Port      Vlans allowed and active in management domain
Gig0/2    none

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/2    none

Switch#show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          : CAMPUS
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0001.9627.ED00
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00
Local updater ID is 0.0.0.0 (no valid interface found)

Feature VLAN :
-----
VTP Operating Mode       : Server
Maximum VLANs supported locally : 255
Number of existing VLANs : 5
Configuration Revision   : 0
MDS digest               : 0xEA 0x87 0x1F 0x3C 0x65 0x51 0x13 0x99
                        : 0xF1 0xE3 0x0D 0x72 0x54 0x44 0xA8 0x23

Switch#
```

92) SW2

```
Switch>ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface f0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface f0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
```

R1:

```
Router#iniena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface f0/0
%Invalid interface type and number
Router(config)#
Router(config)#
Router(config)#interface g0/0
Router(config-if)#ip address 10.10.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface g0/1
Router(config-if)#ip address 10.10.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

94)SW2

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface f0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#no shutdown
Switch(config-if)#interface f0/2
Switch(config-if)#shutdown

Switch(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
```

R1

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no ip address
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface g0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 10.10.10.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface g0/0
Router(config-if)#interface g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 10.10.20.1 255.255.255.0
% 10.10.20.0 overlaps with GigabitEthernet0/1
Router(config-subif)#
```
